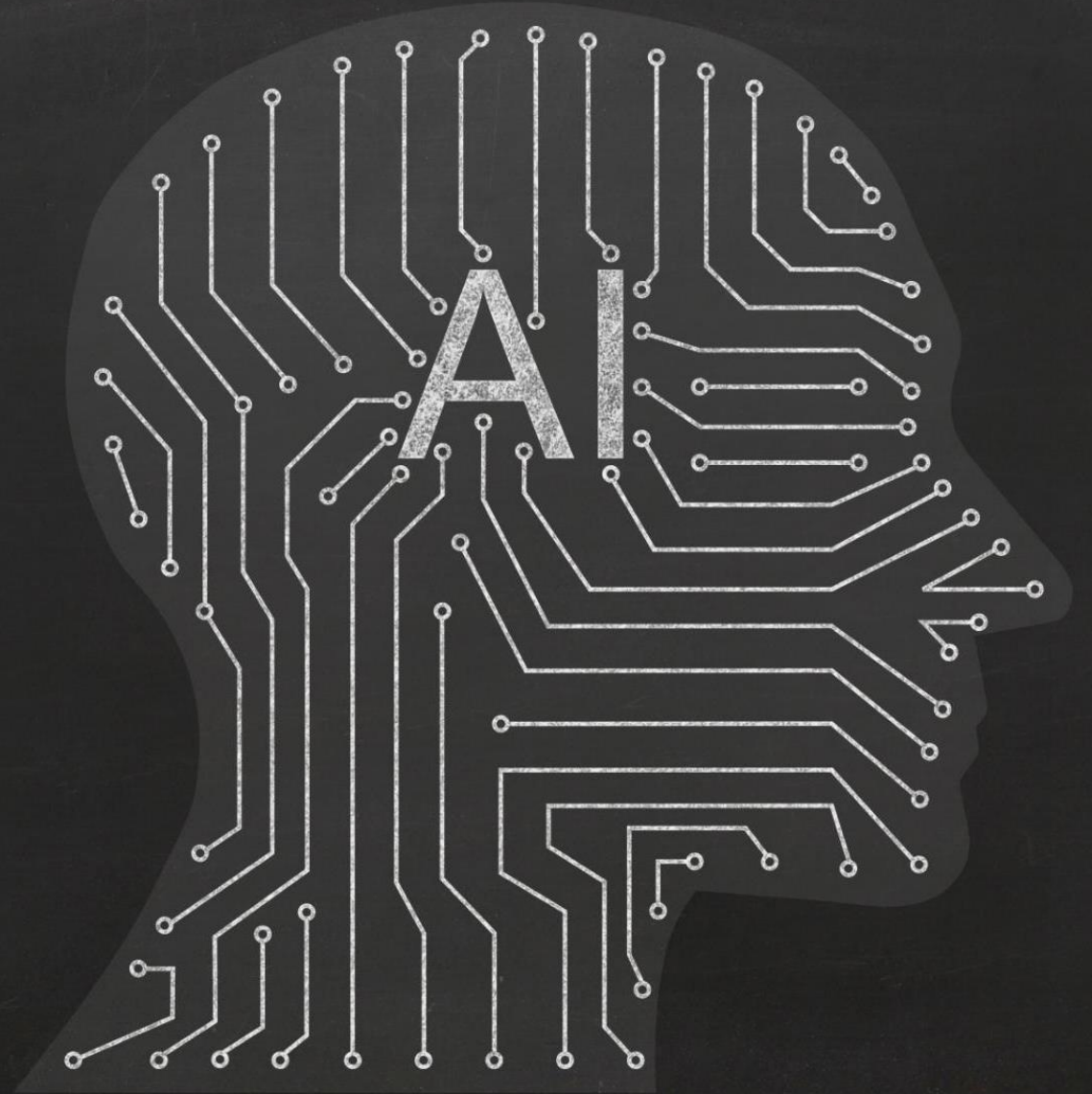


Self- Supervised Learning (SSL)



Course Title: Mechine Learning
Course code: CSE 475
Section:02

Submitted to
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CONTENT

Constructive SSL

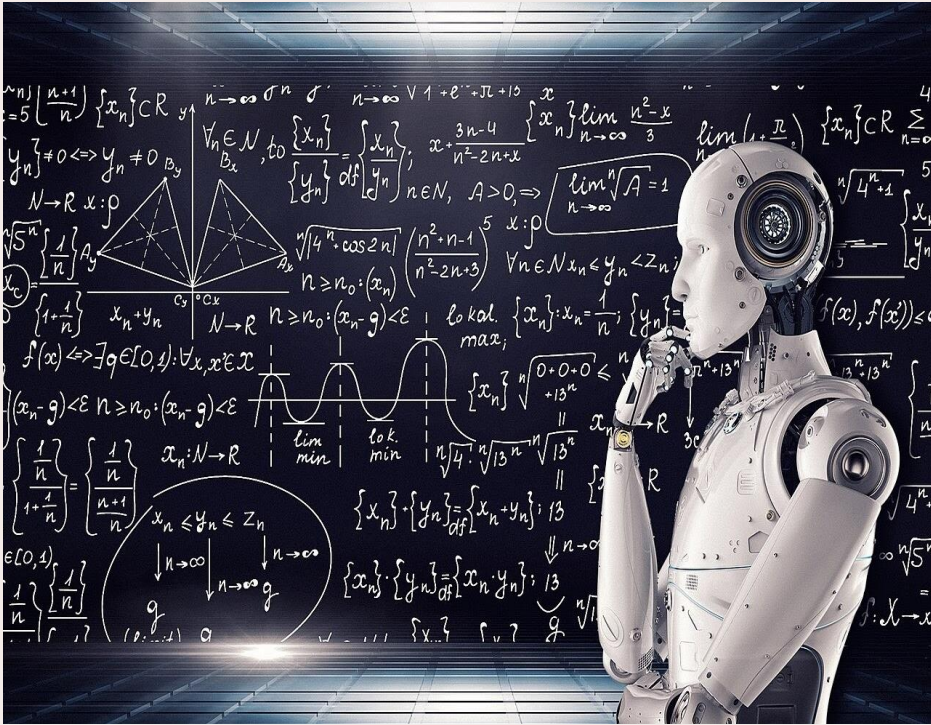
SimCLR(Simple Contrastive Learning Framework)Model

BYOL(Bootstrap Your Own Latent)

Non-Constructive SSL

MAE (Masked Autoencoder) Model

SimCLR



- SimCLR is a self-supervised learning method that learns image features without labels.
- It makes two augmented versions of the same image and trains the model to bring them closer while pushing apart different images.
- This helps the model learn useful visual features for later tasks like classification.

SIMCLR – CONCEPT OF CONTRASTIVE LEARNING

SimCLR is based on similarity learning. It teaches the model which images are alike and which are different.

It uses contrastive loss to pull similar image pairs closer and push dissimilar pairs apart in feature space.

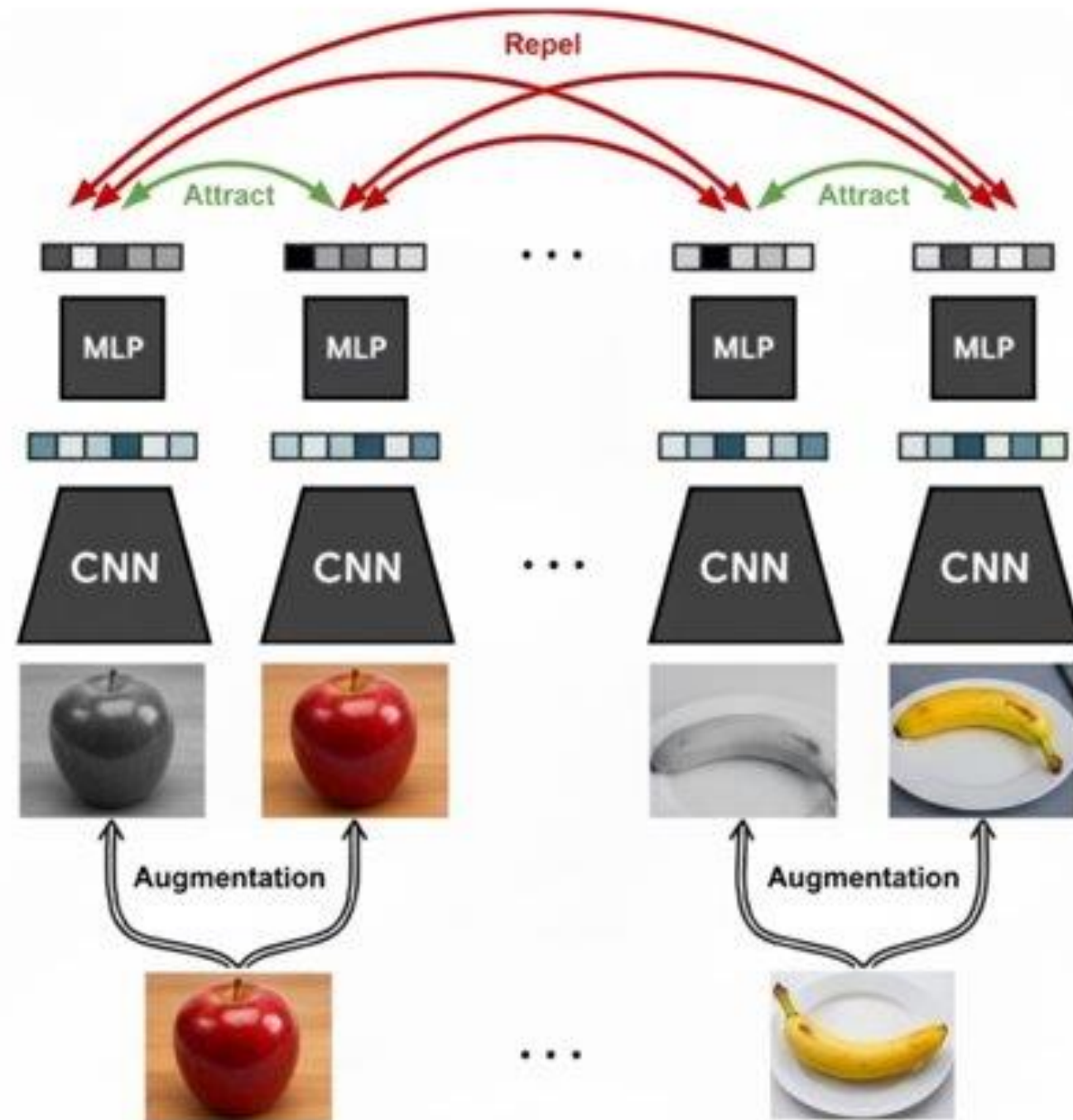
. How It Works

Take an image

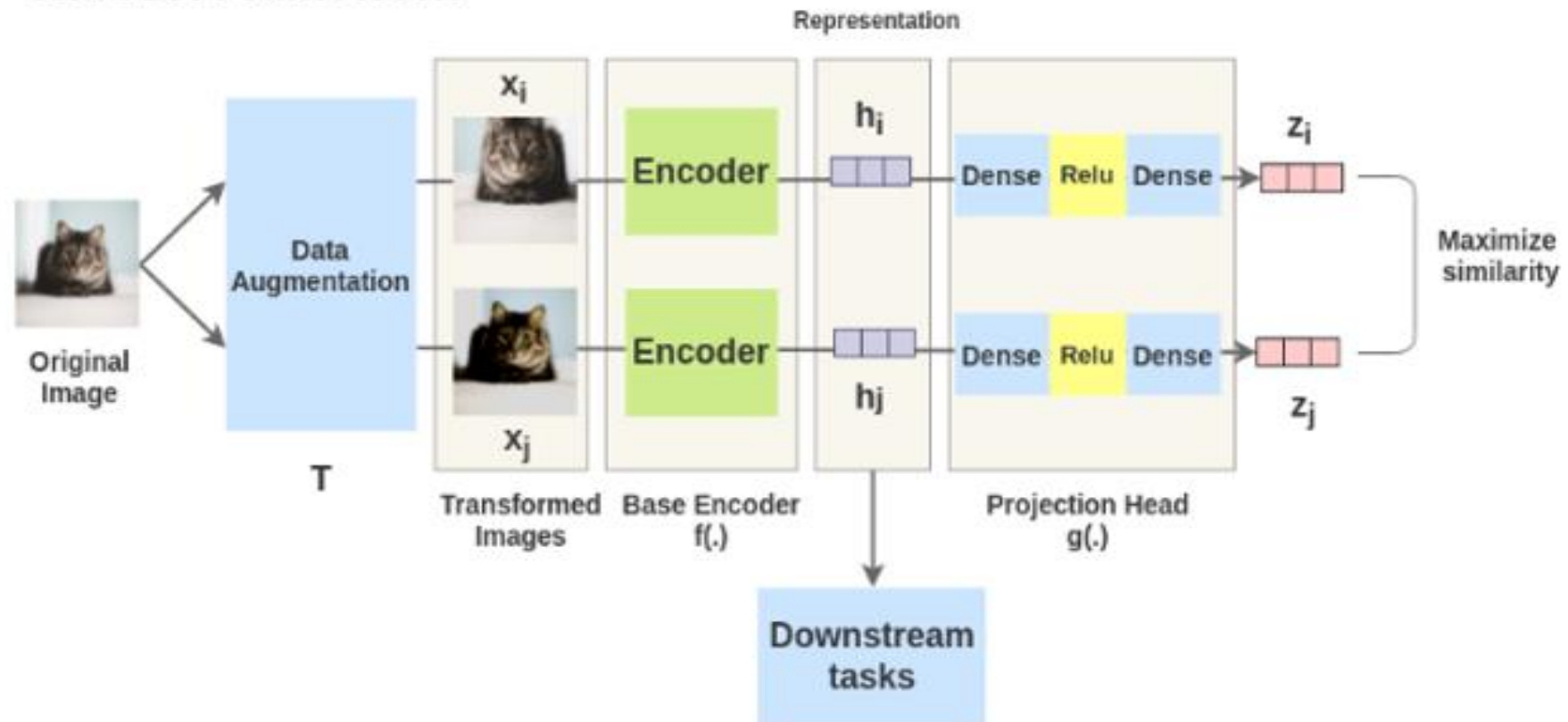
Create two different augmented versions (crop, color change, flip, etc.)

Pass both through the same encoder network (ResNet)

Use contrastive loss to make their embeddings similar



SimCLR Framework



Results

SimCLR achieved high performance on image classification tasks without labels.

When fine-tuned, it performs almost as well as fully supervised models.

Learned representations are transferable to other tasks (detection, segmentation).

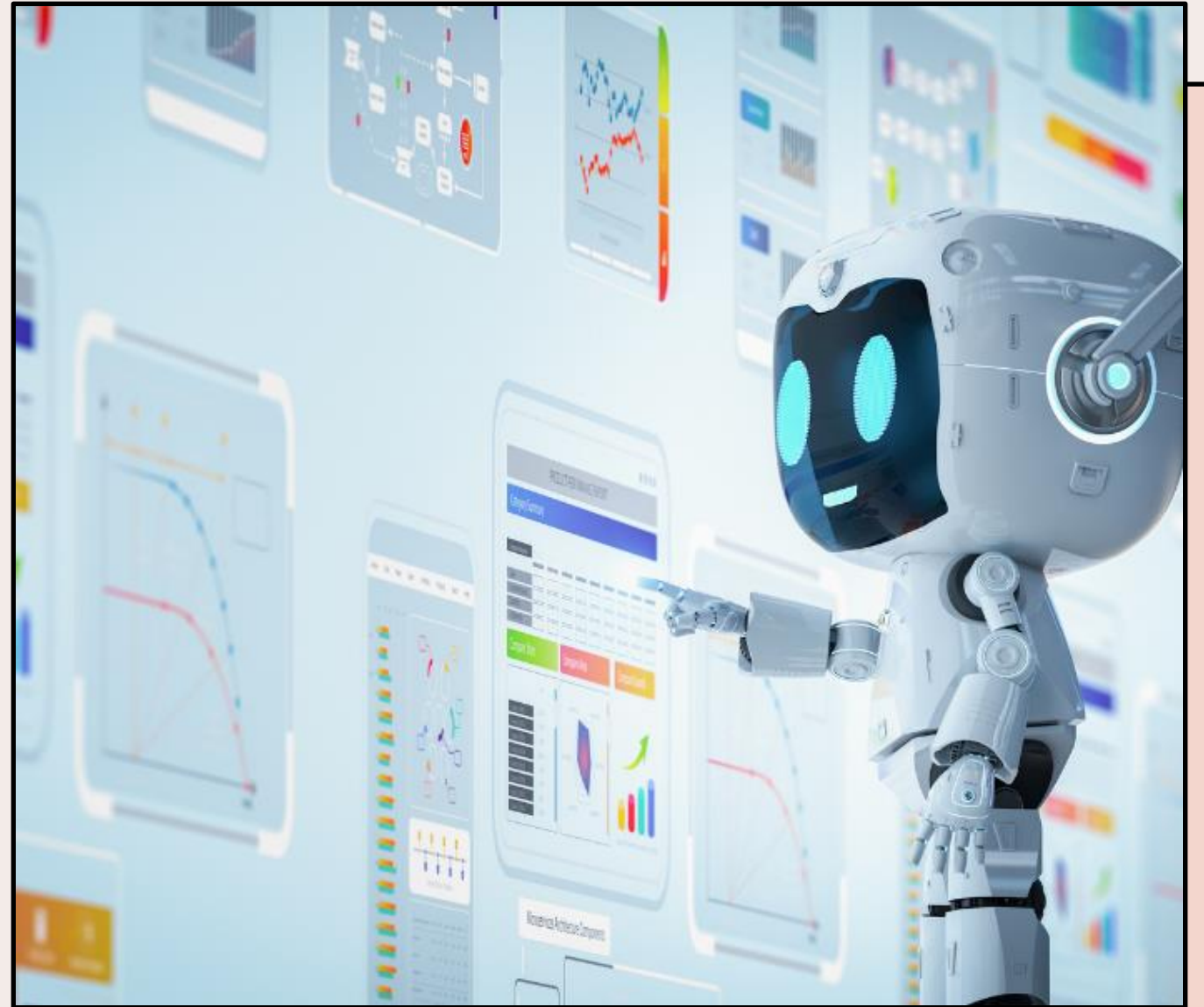
Advantages

No labeled data needed → perfect for large unlabeled datasets.

Learns general and strong visual features.

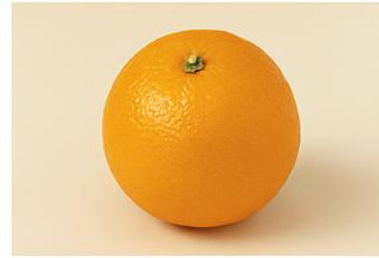
Works well for pretraining, then fine-tuning for classification.

Improves data efficiency and model robustness.

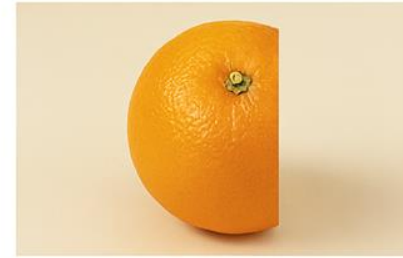


BYOL (Bootstrap Your Own Latent)

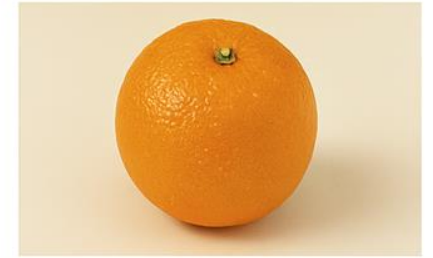
- A Constructive Self-Supervised Learning (SSL) method
- Learns without labeled data
- Uses two networks: Online & Target
- Learns by matching its own representations
- No negative pairs needed (unlike SimCLR)
- Builds strong, transferable feature representations



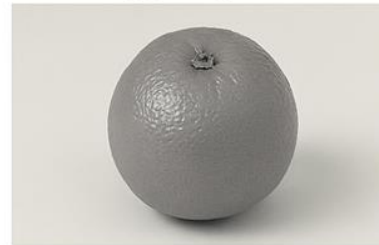
(a) Original



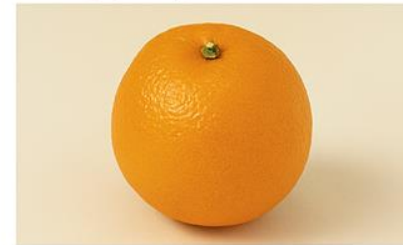
(b) Crop and rotated



(c) Color noised flip



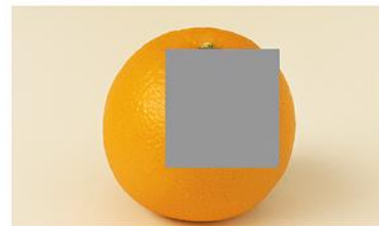
(d) Color distort, c'



(e) Gaussian noise



(f) Rotated



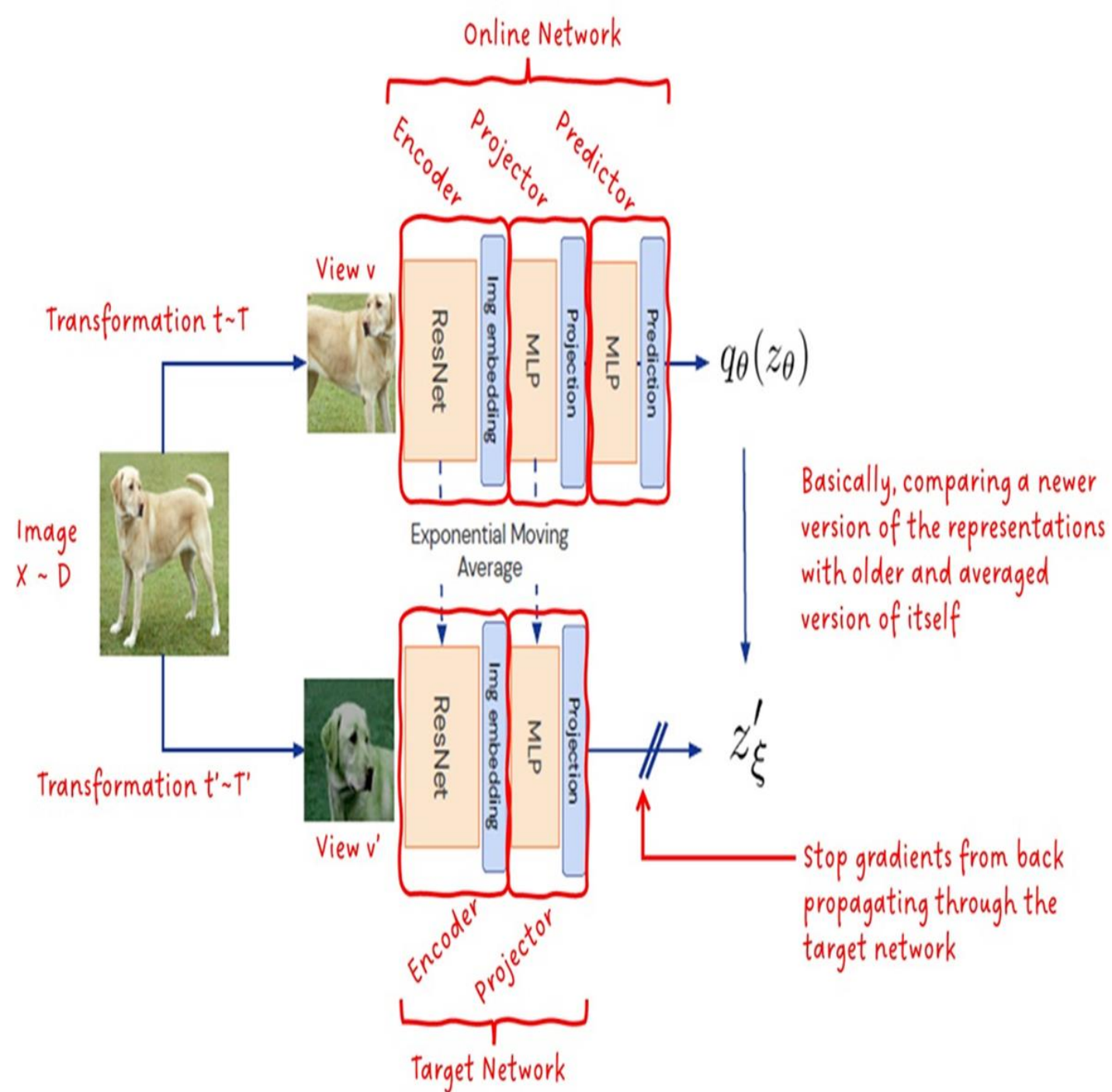
(g) Cutout



(h) Gaussian noise



(i) Sobel filtering



BYOL Architecture

BYOL'S LOSS FUNCTION

$$\mathcal{L}_{\theta,\xi}^{\text{BYOL}} = \mathcal{L}_{\theta,\xi} + \tilde{\mathcal{L}}_{\theta,\xi}$$

Loss for
 $V \rightarrow$ online network
 $V' \rightarrow$ target network

Loss for
 $V \rightarrow$ target network
 $V' \rightarrow$ online network

„Defined as“
operator

$\|a - b\|_2^2$:
squared L2-distance
between a and b

$\mathcal{L}_{\theta,\xi} \triangleq \|\overline{q_\theta}(z_\theta) - \overline{z'_\xi}\|_2^2 = 2 - 2 \cdot \frac{\langle q_\theta(z_\theta), z'_\xi \rangle}{\|q_\theta(z_\theta)\|_2 \cdot \|z'_\xi\|_2}$

L2-normalized output
of online network

L2-normalized output
of target network

$\langle a, b \rangle$: inner product
between a and b

$\|a\|_2$: L2-normalized
vector a

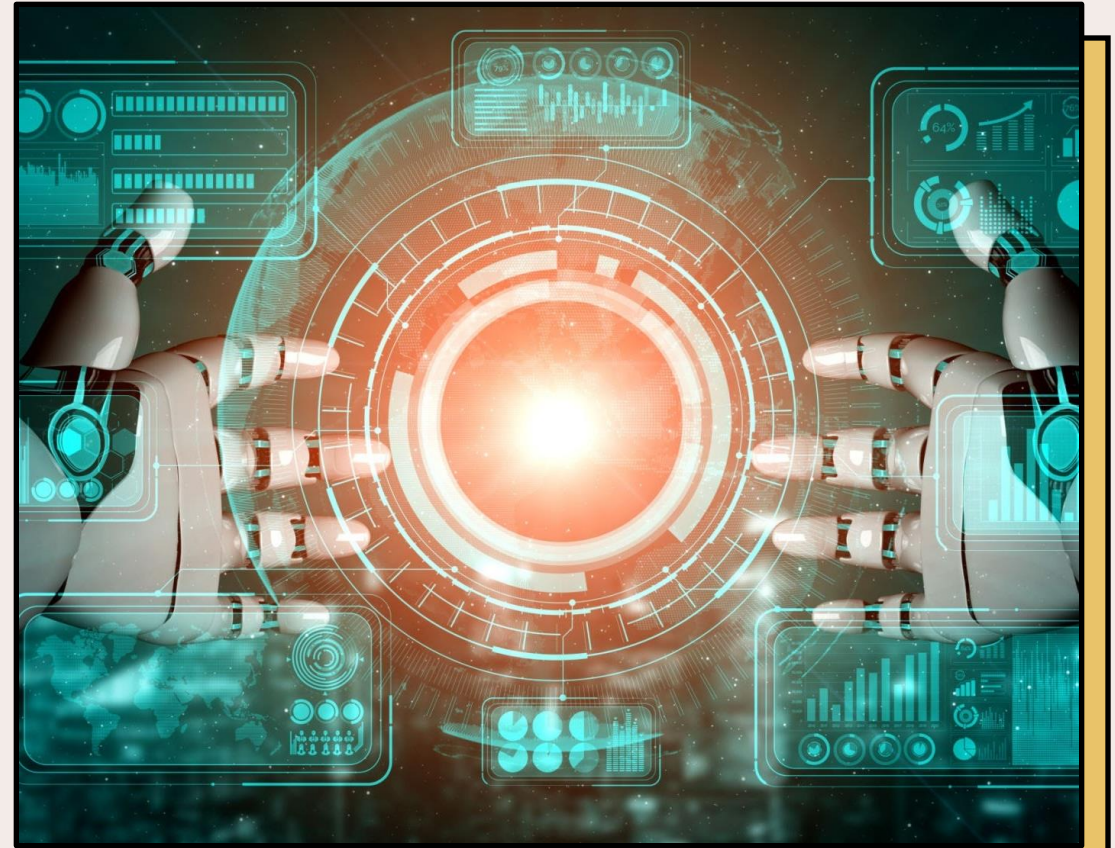
Result And Advantages

Results

- Very good accuracy without using labels
- Works better than SimCLR on many datasets
- Learns features that can be used in other tasks
- Works well with different network types

Advantages

- No labeled data needed
- No negative pairs required
- Easy and stable training
- Learns strong & useful features
- Good for small or unlabeled datasets



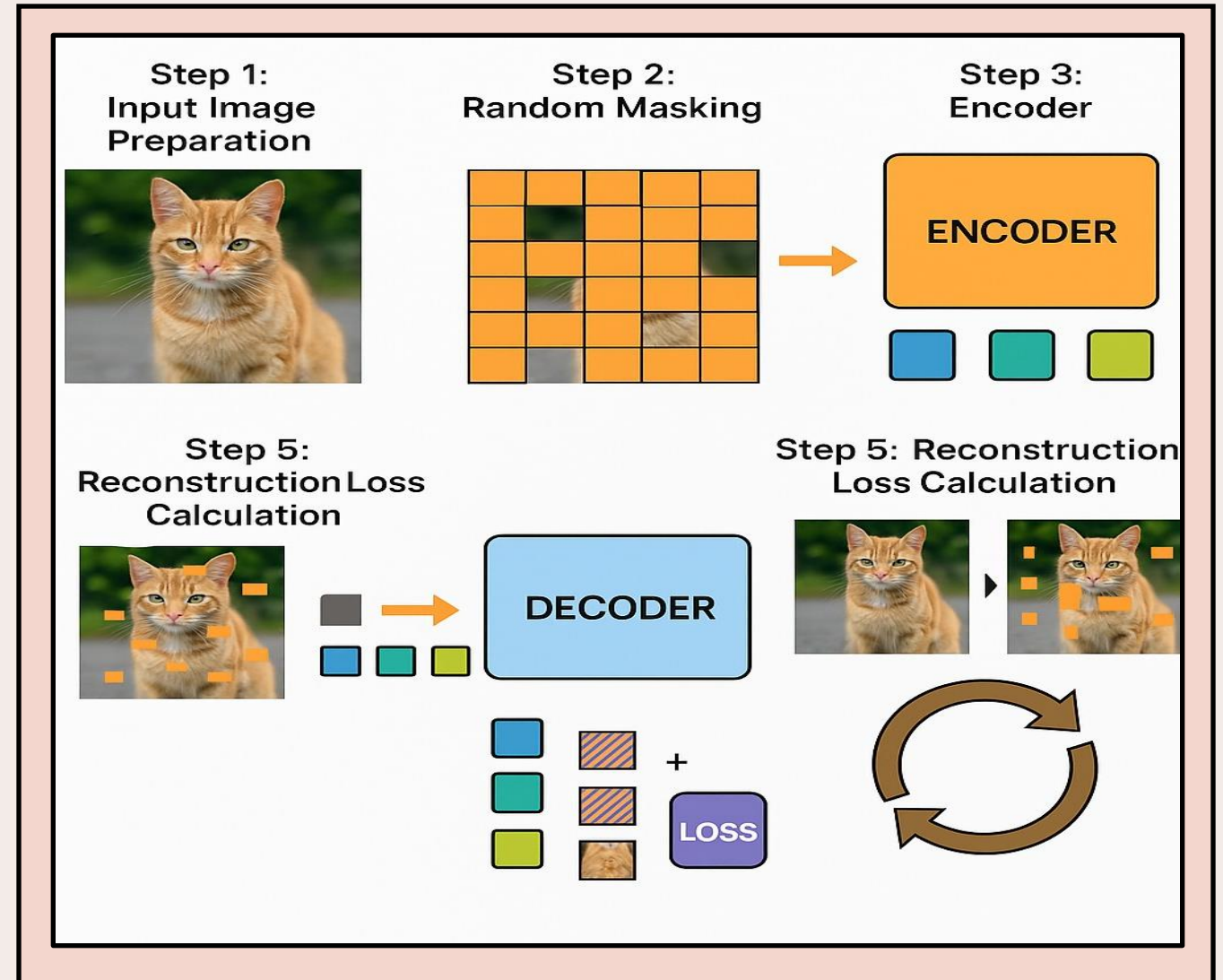
MAE

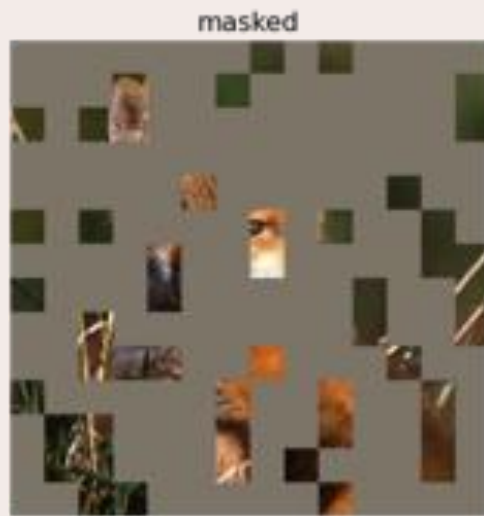
MAE (Masked Autoencoder) is a self-supervised learning method that learns by reconstructing the missing parts of an image.

It does not require labeled data, the model teaches itself by predicting what's hidden.

Overview

- The input image is partially masked (e.g., 75% patches hidden).
- The encoder sees only the visible patches.
- The decoder tries to reconstruct the missing patches.
- The model learns to understand overall image structure and context.





MASKED AUTOENCODER: WORKFLOW OVERVIEW

Step by Step Architecture

Input Image → Split into patches (16×16)
Random Masking → hide 75% of patches
Visible Patches → Encoder (Vision Transformer)
Encoded Latent Representations → Decoder
Reconstructed Image → Compare with original (loss)

Encoder

Processes only visible (unmasked) patches.
Learns meaningful visual features.
Efficient and fast because it ignores masked parts.

Decoder

Takes encoded features + mask positions.
Reconstructs the missing patches.
Used only during pretraining (not in fine-tuning).



Loss Function

Mean Squared Error (MSE) over Masked Patches

$$L = \frac{1}{|M|} \sum_{i \in M} \|x_i - \bar{x}_i\|^2$$

Where:

- M = set of indices of masked patches
- x_i = original pixel values of patch i
- $\bar{x}_i$ = reconstructed pixel values of patch i
- $|x_i - \bar{x}_i|^2$ = squared error for patch i

Purpose:

To measure how well the model reconstructs the masked patches by comparing the original and predicted pixel values.

Advantages

No Labeled Data Needed

- MAE learns directly from raw, unlabeled images.
- Perfect for large datasets where manual labeling is expensive or impossible.

Efficient Representation Learning

- The encoder learns rich and general visual features,
- which can be reused for classification, detection, or segmentation tasks.

Data Efficiency

- Can achieve strong performance even with less labeled data during fine-tuning.

Scalable and Flexible

- Works well with large Vision Transformer (ViT) architectures.
- Can be scaled up easily for bigger datasets.



Thank
youooooooooooooo.

